the point $d_A(\mathbf{x}_{obs})$ to locate within the reference. The consequence is that the PPC might be overconfident about a false model.

Bayarri and Berger (2000) and Robins et al. (2000) examine this issue, both theoretically and empirically. They consider the sampling distribution of the p-value as a function of (random) observations $\mathbf{x}_{obs}$ from a true likelihood $p(\mathbf{x}_{obs} \mid \theta^*)$. A *calibrated* p-value has a uniform sampling distribution when the data truly come from this model. Calibration is necessary to interpret p-values; if we do not know the distribution of p-values under the null hypothesis, we cannot make a decision on whether the p-value is "surprising" or not. Bayarri and Berger (2000) and Robins et al. (2000) show that Eq. 5 is not calibrated.

Bayarri and Berger (2000) also propose alternative reference distributions, called partial posterior predictives, for which the p-values enjoy better calibration. This paper will use an adaptation of their check, which we will refer to as the *heldout predictive check*. The heldout predictive check divides the observed data into two sets $\mathbf{x}_{obs} = (\mathbf{x}_{in}, \mathbf{x}_{out})$, uses $\mathbf{x}_{in}$ to form the reference distribution, and locates $d_A(\mathbf{x}_{out})$ within it. The check is

$$p_{hpred} = P(d_A(\mathbf{x}_{rep}) \geq d_A(\mathbf{x}_{out}) \mid \mathbf{x}_{in}) \qquad \mathbf{x}_{rep} \sim p_A(\mathbf{x}_{rep} \mid \mathbf{x}_{in}). \tag{6}$$

This type of check relates closely to predictive checks that rely on cross-validation (Gelfand et al., 1992; Marshall and Spiegelhalter, 2007) and held-out data (Draper, 1996).

2.2 Posterior predictive null checks

The spirit of a predictive check is to try to falsify a model. If we find an observed diagnostic in the tail of the reference distribution then we "reject the model," taking a p-value as a measure of the risk of falsely rejecting a plausible model. When the observed diagnostic is not in the tail—when it has "passed the check"—then we have not (yet) falsified the model. With this perspective, Gelman and Shalizi (2012) relate PPCs to the Popperian view of the philosophy of science.

There is an important side of model criticism, however, that a predictive check does not address. Suppose model A is not rejected; it passes its PPC. This result means that $d_A(\mathbf{x}_{obs})$ is plausible under the model-A predictive distribution, and we do not reject model A. But does that mean we should accept it?

To help answer this question, we propose the posterior predictive null check (PPN). Consider a different model B and suppose that it provides the same posterior predictive distribution as model A. This means that data from model B will pass the predictive check for model A, i.e., that data from model B can "fool" the check for model A. In this case, we would conclude that model B captures the data equally well as model A (with respect to the chosen diagnostic). This is exactly what the PPN is designed to test. Simply, the PPN asks whether the two models produce the same posterior predictive distribution of the model-A diagnostic. While a predictive check assesses whether the model is adequate, a PPN helps to assess whether the model is necessary to represent the data.

Consider again Figure 1 (Left), which shows two-dimensional data from a mixture of three Gaussians. There are clearly three clusters. Figure 1 (right) shows draws from the corresponding posterior predictive for four models, $K = \{1, 2, 3, 4\}$. As expected, a 3-mixture provides a good posterior predictive distribution but notice that $K = 4$ does as well; it simply splits one of the clusters. The predictive checks corroborate this visual insight—both $K = 3$ and $K = 4$ pass their check, and the partial predictive p-values (Eq. 6) are 0.42 and 0.45, respectively. (In fact, each of these models passes its check.)

A PPN can help assess $K = 4$ by asking when data from the 3-mixture's posterior predictive can fool the check for the 4-mixture. This question amounts to asking if the distribution of the $K = 4$ diagnostic $d_4(\mathbf{x}_{\text{rep}})$ is the same whether $\mathbf{x}_{\text{rep}}$ is drawn from $K = 4$ posterior predictive, which is the reference distribution of its predictive check, or the $K = 3$ posterior predictive. If these two posterior predictive distributions are the same then either one will adequately locate the observed diagnostic $d_4(\mathbf{x}_{\text{obs}})$ in the $K = 4$ reference distribution. Consequently, passing the predictive check for $K = 4$ does not rule out the possibility that the data came from $K = 3$ (which, in this case, it did).

Definition 1 (Posterior predictive null check; PPN)**.** *Consider two models,* A *and* B*, and their posterior predictive distributions. Each model involves its own set of latent variables, but defines a distribution on the same observation space* $\mathbf{x} \in \mathcal{X}$*,*

$$\mathrm{p_A}(\mathbf{x}_{\text{obs}}, \theta_{\mathrm{A}}) = \mathrm{p_A}(\mathbf{x}_{\text{obs}}|\theta_{\mathrm{A}})\mathrm{p_A}(\theta_{\mathrm{A}}) \qquad \mathrm{p_A}(\mathbf{x}_{\text{rep}} \,|\, \mathbf{x}_{\text{obs}}) = \int \mathrm{p_A}(\mathbf{x}_{\text{rep}} \,|\, \theta_{\mathrm{A}})\mathrm{p_A}(\theta_{\mathrm{A}} \,|\, \mathbf{x}_{\text{obs}}) \,\mathrm{d}\theta_{\mathrm{A}}$$

$$\mathrm{p_B}(\mathbf{x}_{\text{obs}}, \theta_{\mathrm{B}}) = \mathrm{p_B}(\mathbf{x}_{\text{obs}}|\theta_{\mathrm{B}})\mathrm{p_B}(\theta_{\mathrm{B}}) \qquad \mathrm{p_B}(\mathbf{x}_{\text{rep}} \,|\, \mathbf{x}_{\text{obs}}) = \int \mathrm{p_B}(\mathbf{x}_{\text{rep}} \,|\, \theta_{\mathrm{B}})\mathrm{p_B}(\theta_{\mathrm{B}} \,|\, \mathbf{x}_{\text{obs}}) \,\mathrm{d}\theta_{\mathrm{B}}.$$

Consider a diagnostic function for model A *denoted by* $d_{\mathrm{A}}(\cdot)$*, such as a residual (Eq. 4), and an observed dataset* $\mathbf{x}_{\text{obs}}$*. The posterior predictive null check* $\mathrm{PPN}(d_{\mathrm{A}}, \mathrm{p_A}, \mathrm{p_B}, \mathbf{x}_{\text{obs}})$ *assesses the similarity between the posterior predictive distributions of the two models under the model-*A *diagnostic. With a divergence,* D*, the PPN is:*

$$\mathrm{PPN}(d_{\mathrm{A}}, \mathrm{p_A}, \mathrm{p_B}, \mathbf{x}_{\text{obs}}) = D(\mathrm{p_A}(d_{\mathrm{A}}(\mathbf{x}_{\text{rep}}) \,|\, \mathbf{x}_{\text{obs}}) \parallel \mathrm{p_B}(d_{\mathrm{A}}(\mathbf{x}_{\text{rep}}) \,|\, \mathbf{x}_{\text{obs}})). \tag{7}$$

One example is the symmetrized Kullback-Leibler divergence:

$$D_{\text{SymKL}}(P \parallel Q) = 0.5 D_{\text{KL}}(P||Q) + 0.5 D_{\text{KL}}(Q||P), \tag{8}$$

where $D_{KL}(P||Q) = \int_{-\infty}^{\infty} p(x) \log[p(x)/q(x)] dx$ is the Kullback-Leibler divergence between distributions P and Q. Another less precise example is visual inspection of the densities $\mathrm{p_A}(d_{\mathrm{A}}(\mathbf{x}^{\mathrm{A}}_{\text{rep}}) \,|\, \mathbf{x}_{\text{obs}})$ and $\mathrm{p_B}(d_{\mathrm{A}}(\mathbf{x}^{\mathrm{B}}_{\text{rep}}) \,|\, \mathbf{x}_{\text{obs}})$.

Return to the Gaussian mixture model, and recall that both $K = 3$ and $K = 4$ passed their respective predictive checks. We use a PPN to check if data from the simpler mixture ($K = 3$) can fool the more complex one ($K = 4$). For the diagnostic $d_4(\cdot)$ we use the Gaussian mixture model likelihood with $K = 4$ components (for further details, see Appendix A of the Supplementary Material). To implement the check, we calculate the empirical distributions of $d_4(\mathbf{x}^{(3)}_{\text{rep}})$ and $d_4(\mathbf{x}^{(4)}_{\text{rep}})$, where $\mathbf{x}^{(3)}_{\text{rep}}$ and $\mathbf{x}^{(4)}_{\text{rep}}$ are draws from the posterior predictive of the 3- and 4-mixtures, respectively.

This analysis is illustrated in the bottom row of Figure 2; all the panels in the row involve the model $K = 4$. In the rightmost panel is an illustration of the partial predictive check. The distribution is the posterior predictive of the diagnostic and the red line is the observed diagnostic (from held-out data); the model $K = 4$ passes its predictive check. The panels to the left illustrate different PPNs, each illustrating the predictive distribution of $\mathrm{p}(d_4(\mathbf{x}_{\mathrm{rep}})|\mathbf{x}_{\mathrm{obs}}, K = 4)$ (blue) and $\mathrm{p}(d_4(\mathbf{x}_{\mathrm{rep}})|\mathbf{x}_{\mathrm{obs}}, K = k)$ for $k = 1, 2, 3$ (red). Specifically, the leftmost panel is a PPN that checks if data from $\mathrm{p}(\mathbf{x}_{\mathrm{rep}} \,|\, \mathbf{x}_{\mathrm{obs}}, K = 1)$ can fool the check for $K = 4$; it cannot. The next panel asks the same question for $K = 2$; again it cannot fool the check. The next panel, however, illustrates the distribution for $K = 3$; data from $\mathrm{p}(\mathbf{x}_{\mathrm{rep}} \,|\, \mathbf{x}_{\mathrm{obs}}, K = 3)$ will pass the check for $K = 4$. Thus we cannot distinguish between the two models. In Appendix B of the Supplementary Material, we compare the models with Bayes factors (Jeffreys, 1961; Kass and Raftery, 1995) and obtain a similar conclusion to the PPN study.

2.3 The PPN study in a Bayesian workflow

How can we incorporate the PPN study in the workflow of Bayesian data analysis (Gelman et al., 2020)? First consider a set of models that are ordered by their natural complexity. The mixture models of Figure 1 are a good example. One approach to using a PPN is to iteratively increase the complexity of the model class, use a predictive check to check each model, and use a PPN to check whether any of the simpler models can fool the check. Based on the principle of parsimony, one can choose the model that passes its predictive check and for which no simpler model can fool it.

Figure 2 demonstrates this analysis for the mixture model. First note that the predictive check does not help determine which K is necessary. The diagonal plots show how each model passes its predictive check, including the trivial model where $K = 1$; the reason is that the estimated variance in the log-likelihood is too large to detect an anomaly between the observed and predictive data. The off-diagonal plots demonstrate the value of the PPN study. They show that no simpler model can fool the check for $K = 3$. However, as we discussed, data from the 3-mixture can fool the check for the 4-mixture. Based on this analysis, the researcher can choose $K = 3$.

Next, we consider classes of different types of models that are not necessarily nested within each other. We suggest using a predictive check to check each model and then use a PPN to check every pair. This process will result in an equivalence class of models that the data cannot distinguish.

Consider again two models A and B and assume that they both pass their respective predictive check. Now consider two PPNs, one to check if data from model B can fool model A and one to check if data from model A can fool model B. There are three possibilities,

- Suppose data from A fools B and data from B fools A. Then these two models are in an equivalence class. Relative to their diagnostics, neither provides information that the other does not. We may use a qualitative criterion to select the model (e.g., parsimony, as we did for mixtures) or hold them both.